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Short-term thinking, long-term costs: How managerial myopia undermines green innovation

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Purpose: This study investigates the impact of managerial myopia on green innovation in Chinese listed companies from 2001 to 2021, focusing on how short-termism undermines sustainable innovation at both firm and industry levels.

Design/methodology/approach: Utilising Management Discussion and Analysis (MD&A) textual analysis and corporate patent data, we employ high-dimensional fixed-effects and robustness tests to assess the relationship between managerial myopia and green innovation. Heterogeneity analysis and mediation models explore variations across firm types and mediating mechanisms, including R&D investment, digital transformation and environmental, social and governance (ESG) management.

Findings/results: Results confirm that managerial myopia significantly inhibits green innovation, with stronger effects in non-state-owned enterprises (non-SOEs), non-heavy polluting firms and non-CEO-duality firms. At the industry level, managerial myopia indirectly suppresses green innovation through reduced R&D investment, limited digital transformation and weaker ESG management, as validated by mediation analysis.

Practical implications: The findings offer policymakers and corporate leaders' insights into aligning managerial incentives with long-term environmental and economic goals. Encouraging sustainable focus in non-SOEs and enhancing R&D and ESG practices could mitigate the adverse effects of short-termism on green innovation.

Originality/value: This research integrates MD&A textual analysis with patent-based metrics to provide a novel perspective on managerial myopia's impact on green innovation in an emerging market context. It elucidates nuanced mechanisms – R&D, digital transformation and ESG management – through which short-termism disrupts sustainability, contributing to the literature on corporate governance and environmental innovation.

Keywords: managerial myopia; green innovation; digital transformation; ESG management; R&D investment.

Introduction

Environmental, social and governance (ESG) performance has become an integral part of corporate value. Against the backdrop of the emphasis on low-carbon and green development, companies are increasingly recognising green innovation issues as not merely a fulfilment of their social responsibilities but also a strategic imperative to enhance their economic interests. Green innovation has the potential not only to improve the environment and enhance corporate social responsibility (Chen, 2008) but also to reduce costs, boost green production efficiency (Chen et al., 2006; Wu et al., 2022) and even unlock business opportunities and market recognition (Chen et al., 2022; Tan & Zhu, 2022). Indeed, the promotion of green innovation has become a necessary measure for companies to remain competitive in the future. Meanwhile, promoting green innovation in firms has become a subject of considerable research interest among scholars (Amore & Bennedsen, 2016; Schiederig et al., 2012; Takalo et al., 2021).

Nevertheless, this study found that managerial myopia can have a negative impact on a company's green innovation from a patent perspective. Managerial myopia refers to the behaviour of company managers who prioritise short-term profit maximisation over long-term strategic development.

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Note: Additional supporting information may be found in the online version of this article as Online Appendix 1.

Managerial myopia is increasingly common in today's business world, but short-term gains often lead to long-term negative consequences, which has been criticised by the academic community (Denis, 2019; Willey, 2019). The existing literature has confirmed that managerial myopia does indeed have a negative impact on long-term investment decisions (Garel, 2017; Li et al., 2019, 2021; Ridge et al., 2014; Yuan et al., 2023), future firm performance (Brochet et al., 2015; Jiang & Xin, 2022; Zhao et al., 2012), production efficiency (Sheng et al., 2022; Willey 2019), resource allocation (Cannon et al., 2020; Chatjuthamard et al., 2023) and other areas.

Although there is a large body of literature on managerial myopia, there is currently little research on its impact on corporate green innovation, especially from the perspective of patent activity. As part of corporate innovation, green innovation is more socially responsible and contributes to ESG value creation than other forms of innovation. From a business perspective, the drivers of green innovation appear to be more directly related to ESG benefits (Tan & Zhu 2022). Indeed, as an important aspect of ESG valuation, a company's green image and sustainable behaviours have become the main frame of reference for management when designing business plans. This article examines the impact of managerial myopia on corporate green innovation and investigates the potential negative effects of managerial myopia. The resulting research findings will provide recommendations and support for addressing the negative effects of managerial myopia and for managing ESG in companies.

This study's significance lies in its pioneering effort to bridge the gap between managerial myopia and green innovation, an area underexplored despite its critical implications for sustainable development. By leveraging patent data as a tangible measure of green innovation, it extends prior work, such as that of Berrone et al. (2013), who highlight the role of environmental innovation in firm's competitiveness, and complements studies such as Flammer (2015), which demonstrate how long-term orientation enhances corporate social performance. In an era where global economies face mounting pressure to transition to low-carbon systems, understanding how managerial short-termism undermines green innovation offers actionable insights for aligning corporate strategies with broader societal goals. This research thus not only enriches the theoretical discourse on managerial behaviour and innovation but also provides a practical framework for policymakers and executives to foster sustainable value creation.

Literature review and research hypotheses

Managerial myopia

Managerial myopia (or short-termism) refers to the tendency of top managers to make decisions that sacrifice the long-term interests of shareholders in order to achieve short-term gains (Narayanan, 1985). Although considered a negative behaviour, managerial myopia represents a type of economic decision that is consistent with market incentives (Stein, 1988, 1989). Existing literature suggests that managerial myopia may be influenced by takeover threats (Cannon et al., 2022; Chatjuthamard et al., 2023; Denis, 2019; Denlertchaikul et al., 2022; Zhao et al., 2012), institutional investors (Li et al., 2019; Wahal & McConnell, 2000), incentive schemes (Edmans et al., 2022; Kraft et al., 2018; Laux, 2012; Laverty, 2004) and shareholder pressure (Hughes, 2014; Rieg, 2015; Samuel, 2000). Managerial myopia has become an important part of shareholder value maximisation research because it can pose difficult-to-measure risks to a firm's current production, future planning and potential business opportunities.

Green innovation and influencing factors

Green innovation encompasses technological advancements that offer substantial benefits in environmental protection and resource utilisation, extending the scope of traditional innovation. Yet, green technology innovation carries new meanings and features distinct from those of traditional technological innovation. Corporate green innovation is influenced by numerous factors, with existing literature highlighting the role of financial development (Chen et al., 2021; Lv et al., 2021), environmental policy (Fan et al., 2022), shareholder pressure (Singh et al., 2022), green credit (Hu et al., 2021a; Liu et al., 2021; Wang et al., 2022), market demand, regulatory pressure (Du et al., 2022), government relations (Zhang et al., 2022) and so on. In fact, there are both subjective and objective drivers of enterprise green technology innovation as a market activity itself. The promotion of corporate green innovation is driven by environmental regulations imposed by the media, government and society on the production and development of enterprises in the new economy (Zhong & Peng, 2022) on the one hand and the spontaneous interests of enterprises for their own sustainable development, ESG assessment and production efficiency on the other hand (Tu & Wu, 2021). This article finds that managerial short-termism can also impact corporate green innovation from a patent perspective, and that there are multiple pathways of influence.

Managerial myopia and green innovation

It is often assumed that there is a causal relationship between managerial myopia and green innovation, but the relevant research is not sufficient. Managerial myopia may affect firms' long-term investment decisions, social responsibility obligations and exploitation of business opportunities (Brochet et al., 2015; Li et al., 2021; Jiang & Xin 2022; Yuan et al., 2023). Meanwhile, these negative effects are directly related to green innovation. The existing extensive literature also indicates that corporate investment in green innovation is a long-term investment, a social obligation and a future business opportunity (Chen, 2008; Chen et al., 2006, 2008; Tan & Zhu 2022; Wu et al., 2022). It is known that R&D investment in long-term corporate investment can effectively promote green innovation (Xu et al., 2021). At the level of exploiting business opportunities, existing studies show that sticking to traditional business and rejecting digital

transformation inhibits green innovation (Song et al., 2022), and managerial myopia affects digital transformation (Li & Zhai, 2022). Therefore, this article argues that green innovation in the digital economy is affected by managerial short-sightedness and that digital transformation may be a mediating factor. At the level of CSR obligations, green innovation is subject to corporate responsibility indicators represented by ESG (Tan & Zhu 2022), and there is also some direct influence of management on ESG (Gillan et al., 2021). Considering the evidence from the above literature, it is certain that management myopia may indeed inhibit corporate green innovation indirectly through the effects of long-term investments, social obligations and business opportunities and may also have some direct effects from the perspective of business decisions and innovation drivers. Therefore, it is reasonable to assume that managerial myopia may influence corporate green innovation and that there may be some mediating effect between them. There is little existing research in the literature, and this article aims to provide a preliminary exploration of this area. This article attempts to explore the impact of management myopia on green innovation from the perspective of management myopia research, and the findings will provide some reference value to the existing impact of management myopia and corporate ESG management.

Research hypothesis

Based on the above research, this article proposes the following hypotheses:

H1: Managerial myopia has a disincentive effect on corporate green innovation.

H2a: Managerial myopia indirectly inhibits green innovation through a weakening of R&D investment.

H2b: Managerial myopia indirectly inhibits green innovation through resistance to digital transformation.

H2c: Managerial myopia indirectly inhibits corporate green innovation through the weakening of ESG management.

This is the logical framework diagram for this article (Figure 1).

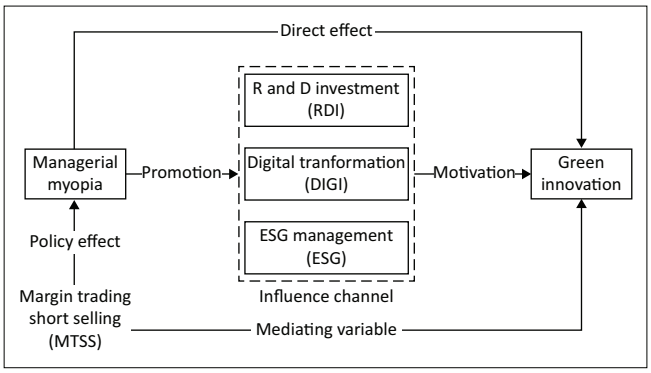

FIGURE 1: Main logical framework.

Methodology

Data sources

The core explanatory variable of this article, management myopia, is a difficult economic behaviour to quantify. Most of the traditional literature uses questionnaire scores or R&D inputs to quantify the level of management myopia (Marginson & McAulay, 2008; Bushee, 1998), but unfortunately these two proxy variables have limited descriptive effect. Questionnaires are highly subjective and manipulable, and R&D inputs inherently have short-term items (Laverty, 2004), making these two variables less than ideal proxies for managerial myopia. However, today's sophisticated methods of text analysis are able to describe and analyse them more precisely and quantitatively at a linguistic level.

It is well known that language and word usage can reflect the cognitive, preferential and behavioural characteristics of subjects (Webb et al., 1966), and researchers can trace many characteristics of subjects by analysing the word types and frequency of their linguistic text content (Brochet et al., 2015; Miller & Ross, 1975; Pennebaker et al., 2003). If the expression emphasises words such as 'once upon a time' and 'in the past', it reflects a greater focus on the past, and conversely if the speaker emphasises words such as 'in the future', 'may', 'to go', etc. An earlier study by Brochet et al. (2015) used this mode of analysis to examine managerial myopia when they examined conference calls as a channel for voluntary disclosure and created a proxy for the timeframe that executives emphasised in their communications. The study showed that the timeframe of conference call narratives can provide information about managerial myopic behaviour. Similar analytical models of managerial myopia have been used more frequently, and this article fully incorporates the experience of previous studies (Brochet et al., 2015; Guo et al., 2023a; Hu et al., 2021b; Li, 2010; Sheng et al., 2022) as a way to more accurately portray managerial myopia as an indicator using big data and textual analysis methods. In this article, we use the MD&A content of 4742 A-share listed companies in China from 2001 to 2021 as the research object, determine the Chinese 'short-term horizon' word set through text analysis and machine learning techniques and then use lexicographical methods to construct the managerial myopia indicator. The word set criteria for the short-sightedness indicator were based on Hu et al. (2021b) (Online Appendix 1 Table B1 for the specific criteria), and the seed words for 'short-sightedness' were first identified by combining the existing English 'short-sightedness' word set and the characteristics of the Chinese corpus of MD&A. The Word2Vec machine learning technique was used to obtain an expanded set of words to represent 'short-term horizon' in the financial context. By further filtering the word set, the final set of words for the short-termism of managers of Chinese listed companies was identified. The specific formula for the management myopia index is as follows (Equation 1):

$$Myopia_{(c,\,i,\,t)} = \frac{\text{Total word frequency for short-term perspective vocabulary}}{\text{Total word frequence for MD \& A full text}} * 100 \qquad \text{[Eqn 1]}$$

Considering the time lag of the patent application process itself, the explanatory variable of this article, the green innovation of enterprises, uses the number of green invention applications, the number of green invention licences and the sum of both as proxies in the current year. The data are obtained from the State Intellectual Property Office of China and the CSMAR database and filtered using the WIPO (World Intellectual Property Organisation) list. It should be observed that the sum of the two is taken as the overall level of corporate green innovation, which is supplemented by the respective data for verification. Much of the literature uses the number of patents granted for green inventions as an indicator of green innovation by enterprises in the current year, which is unscientific. Taking China's national patent law as an example, Article 34 of the *Patent Law of the People's Republic of China* stipulates that after receiving a patent application for an invention, the Patent Administration Department of the State Council shall, after a preliminary examination, publish it within 18 months from the date of application if it meets the requirements of this law. Even if the patent invention is processed under the accelerated procedure, it still takes 1 year, so this article strictly takes the number of green invention applications, the number of green invention licences and the sum of the two as proxy variables. For the sake of data rigour, this article also uses the number of green inventions granted with a lag of one-period for comparison.

The mediating variables in this article are R&D investment, digital transformation and company ESG score. As a proxy variable for R&D investment (RDI), this article chooses a company's annual R&D investment funding.

In terms of digital transformation, this article refers to the research results of Wu et al. (2021), Zhao et al. (2021) and Guo et al. (2023b) based on the management discussions and analyses (MD&A) section of the annual report and uses the usage rate of enterprises' digital transformation word frequency as a proxy variable for digital transformation of listed enterprises. Wu et al. (2021) quantify the level of enterprise digital transformation based on five dimensions: artificial intelligence technology, big data technology, cloud computing technology, blockchain technology and digital technology application. Zhao et al. (2021) quantified the digital transformation level of enterprises based on four dimensions: digital technology application, Internet business model, smart manufacturing and modern information system. Although the above two dimensions of digital transformation criteria are different, both are considered to be reliable classification criteria. For the sake of rigour, the sum of the two is used as a proxy variable for the digital transformation (DIGI) of enterprises. The larger the digital transformation indicator, the higher the degree of digital transformation of the company. At the level of enterprise ESG indicators, this article uses the local Chinese ESG indicators (Hua Zheng ESG) as a proxy variable. There are nine levels of ESG ratings, which are assigned from low to high 1-9 according to previous research experience (Feng et al., 2022). The higher the ESG rating, the higher the ESG level of the enterprise.

The choice of control variables is based on a large body of important industry literature, and the following control variables are used: firm size (scale), firm leverage (leverage), annual net profit (net_profit), proportion of independent directors (inde_director), firm age (age), duality (ceo_duality), cash ratio (cash_ratio), board size (board_size), fixed asset ratio (fix_ratio), return on total assets (roa), firm nature (SOE) and firm pollution (heavy_pollut). If there is a CEO duality situation, ceo_duality is assigned 1; otherwise it is 0. In the enterprise management, ceo_duality is assigned 1; otherwise it is 0. If the enterprise is a state-owned enterprise, SOE is 1, otherwise 0. If the enterprise is a heavy polluter, heavy_pollut is 1, otherwise 0. Data for the control variables were mainly obtained from the CSMAR and WIND financial databases. The criteria for classifying heavily polluting enterprises in this article are are based on the Ministry of Environmental Protection of China's Environmental Verification Industry Classification Management List for listed companies.

A description of the data for the above key variables is shown in Table 1.

Empirical model

Baseline model

As the causal relationship explored in this article involves a large number of variables, a more scientific high-dimensional fixed-effects model is used for empirical testing to better address the endogeneity issue. High-dimensional fixed-effects can more comprehensively control for the unobservable heterogeneity that is unique to each individual, thus effectively avoiding, to some extent, the adverse effects on causal inference caused by omitted variable bias and so on. The high-dimensional fixed-effects estimation method proposed by Correia (2016) can not only more effectively control for high-dimensional fixed-effects

TABLE 1: Data description.

Variable	Log	Observation	Mean	SD	Min	Max
Myopia	YES	48112	0.74	0.04	0.690	1.27
green_grant	YES	48112	0.20	0.58	0.000	6.58
green_apply	YES	48112	0.42	0.87	0.000	7.78
green_sum	YES	48112	0.50	0.95	0.000	7.79
Rdi	YES	30214	8.51	1.60	0.000	15.81
Digi	YES	48112	2.62	1.42	0.000	7.31
ESG	YES	36769	2.00	0.16	0.000	2.30
Scale	YES	48112	3.96	1.10	0.690	10.16
Leverage	YES	48112	3.68	0.60	1.833	4.75
net_profit	YES	48112	2.90	0.41	0.650	4.68
cash_ratio	YES	48112	3.85	1.26	0.000	7.06
fix_ratio	YES	48106	0.19	0.14	0.000	0.68
Roa	YES	48106	0.71	0.04	0.480	0.79
inde_director	NO	48042	0.35	0.10	0.000	1.00
Age	NO	48112	9.73	6.89	0.000	32.00
ceo_duality	NO	48112	0.25	0.43	0.000	1.00
board_size	NO	48112	8.41	2.22	0.000	22.00
SOE	NO	50246	0.41	0.49	0.000	1.00

SD, standard deviation; SOE, state-owned enterprises; ESG, environmental, social and governance; RDI, research & development investment; digi, digital transformation level; ESG, environmental, social and governance; ROA, return on total assets; SOE, state-owned enterprises.

in large sample sets but also allows for standard deviation (SD) clustering analysis. Therefore, this estimation method is mainly used in this article to examine the effect of managerial short-termism on green innovation. In addition, high-dimensional fixed-effects estimation can be extended to other models using the Frisch-Waugh-Lovell (FWL) theorem (Frisch & Waugh, 1933), such as instrumental variable (IV) estimation with two-stage least squares (2SLS), which will help to overcome possible endogeneity problems in high-dimensional fixed-effects models. Therefore, this article also tests the robustness of the model using the 2SLS method. Specifically, to investigate the mechanism of managerial short-termism on green innovation, the following econometric model is developed in this article (Equation 2):

$$In(Green_x)_{cit} = \alpha_0 + \alpha_1 In(Myopia)_{cit} + \alpha_2 Contol_{cit} + \mu_c + \delta_i + \tau_t + \varepsilon_{cit} \quad \text{[Eqn 2]}$$

where c represents the city, i represents the industry or company and t represents the year. *Green_x* is the explained variable, indicating the green innovation performance (GI) of the company or industry i in city c, year t. There are three variables representing the level of GI: the number of green patent applications (Green_apply), the number of green patent invention patents granted (Green_grant) and the total number of green inventions and innovations (Green_sum). *Myopia* is the core explanatory variable, which denotes the level of managerial myopia calculated according to text analysis method. *Mediator* is represented by three variables: R&D investment (rdi), digital transformation (digi) and ESG level (esg). *Contol* is characterised by a series of firm variables. We account for the unique context of Chinese listed companies, which often experience events such as shell transactions, strategic relocations and industry shifts, as well as the varying strategic emphasis on green development across different cities. To address these factors and reduce potential omitted variable bias, we incorporate firm, industry and city fixed-effects. μ_i, δ_t and τ_t are city fixed-effects, industry or company fixed-effects and time fixed-effects, respectively. ϵ_{cit} is the random error term.

Equation 2 is a static panel model that examines managerial myopia on green innovation. Considering that green innovation may have certain path-dependent characteristics for itself, in other words, the level of green innovation in the early stage may have an impact on the green innovation outcome in the later stage. Therefore, this article adds the first-order lag term of green innovation (GI) to Equation 2 to establish a dynamic panel model. The econometric model is as follows (Equation 3):

$$In(Green_x)_{cit} = \beta_0 + \beta_1 In(Green_x)_{ci,\ t-1} + \beta_2 In(Myopia)_{cit} + \beta_3 Control_{cit} + \mu_c + \delta_i + \tau_t + \varepsilon_{cit} \quad \text{[Eqn 3]}$$

where $In(Green_x)_{ci,\ t-1}$ is the first-order lag term of $In(Green_x)_{cit}$. Other variables are defined as above.

Mediated effect model

According to the previous hypothesis H2, managerial myopia may also have an impact on green innovation in a firm or industry through three pathways: research and development investment (rdi), digital transformation (digi), and ESG management (esg). To test whether the above variables can play a significant mediating role in management short-sightedness, this article uses a mediating effects model combined with multidimensional fixed-effects estimation to test the mechanism. This section still uses the more robust dynamic panel model, which is constructed as follows:

$$In(Mediator)_{cit} = \gamma_0 + \gamma_1 In(Mediator)_{ci,\ t-1} + \gamma_2 In(Myopia)_{cit} + \gamma_3 Contol_{cit} + \mu_c + \delta_i + \tau_t + \varepsilon_{cit} \quad \text{[Eqn 4]}$$

$$In(Green_x)_{cit} = \varphi_0 + \varphi_1 In(Green_x)_{ci,\ t-1} + \varphi_2 In(Myopia)_{cit} + \varphi_3 In(Mediator)_{cit} + \varphi_4 Contol_{cit} + \mu_c + \delta_i + \tau_t + \varepsilon_{cit} \quad \text{[Eqn 5]}$$

where *Mediator* denotes the mediating variable and the other variables are set as described above. Equation 4 re-tests the relationship between management short-sightedness (*Myopia*) and the mediating variable *Mediator*. Equation 5 verifies the relationship between managerial myopia (*Myopia*) and green innovation *Green_x* after controlling for the mediating variables in the regression model.

Ethical considerations

This article does not contain any studies involving human participants performed by any of the authors.

Results

High-dimensional fixed-effects benchmark results – Static panel

The results in Box 1 explain the static panel results of the high-dimensional fixed-effects of management short-termism on green innovation, where regressions (1), (2), (5) and (6) are baseline regression results without control variables, and (3), (4), (7) and (8) are baseline regression results with control variables. Also, the odd-term regressions are industry fixed effects, while the even-term regressions are firm fixed effects accordingly. The results show that management short-sightedness has a significant inhibitory effect on both green patent applications and green patent grants, and results controlling for the industry fixed-effects are larger and more significant than the firm level effects. However, the significance and influence coefficient of green patent applications are greater than that of green patent grants. As mentioned earlier, there is a delay of more than 1 year in patent applications. We reasonably guess that in addition to the difference in quantity, it may also be because of the lag effect of patent granting. Therefore, we pass a regression on the sum of the two and a lagged one-period explanatory variable as a robustness test.

Box 2 presents the results of two comparative regression tests. The first is the result of a high-dimensional fixed-effects regression of managerial myopia on the overall level of green

BOX 1: Baseline regressions 1.

List	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables			**Green_apply**				**Green_grant**	
Myopia	-0.64***	-0.26***	-0.52***	-0.16*	-0.24***	-0.09	-0.22**	-0.07
	(-5.61)	(-2.97)	(-4.45)	(-1.91)	(-2.61)	(-1.40)	(-2.266)	(-1.18)
Scale	-	-	0.27***	0.16***	-	-	0.16***	0.08***
	-	-	(12.15)	(11.59)	-	-	(8.40)	(7.35)
Leverage	-	-	0.17***	0.09***	-	-	0.08***	0.03**
	-	-	(10.22)	(5.57)	-	-	(7.24)	(2.25)
net_profit	-	-	0.21***	0.10***	-	-	0.14***	0.04**
	-	-	(4.75)	(5.01)	-	-	(4.59)	(2.46)
inde_director	-	-	0.05	-0.03	-	-	0.10	-0.02
	-	-	(0.47)	(-0.61)	-	-	(1.17)	(-0.41)
Age	-	-	0.01***	0.01**	-	-	0.00***	0.01***
	-	-	(4.35)	(2.46)	-	-	(4.32)	(3.61)
ceo_duality	-	-	-0.00	0.01	-	-	-0.01	-0.01
	-	-	(-0.22)	(0.41)	-	-	(-0.77)	(-0.88)
cash_ratio	-	-	0.02***	-0.00	-	-	0.01**	-0.00
	-	-	(3.03)	(-0.08)	-	-	(2.24)	(-0.73)
board_size	-	-	0.00	0.01**	-	-	0.00	0.00
	-	-	(0.82)	(2.07)	-	-	(0.31)	(0.55)
fix_ratio	-	-	-0.25***	0.25***	-	-	-0.11**	1.00***
	-	-	(-3.37)	(3.58)	-	-	(-2.25)	(3.99)
Roa	-	-	-1.41***	-0.58***	-	-	-1.18***	-0.44***
	-	-	(-4.34)	(-4.81)	-	-	(-4.47)	(-5.54)
Property	-	-	0.04	0.00	-	-	0.02	0.00
	-	-	(1.41)	(.)	-	-	(1.37)	(.)
_cons	0.90***	0.61***	-0.64***	-0.7***	0.38***	0.27***	-0.22**	-0.08
	(10.60)	(9.60)	(-3.87)	(-3.23)	(5.57)	(5.66)	(-2.03)	(-0.80)
Cluster City	YES	YES	YES	YES	YES	YES	YES	YES
Company FEs	NO	YES	NO	YES	NO	YES	NO	YES
Industry FEs	YES	NO	YES	NO	YES	NO	YES	NO
City FEs	YES	YES	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES	YES	YES
N	48102	47598	48026	47527	48102	47598	48026	47527
R^2	0.20	0.63	0.32	0.64	0.15	0.58	0.24	0.58

Note: This Box 1 reports the result of least-square regression of managerial short-termism on green innovation and other control variables. Variables are defined in Table 1. Constant shows the regression constant. Company FEs, industry FEs, city FEs and year FEs demonstrates whether company fixed-effects, industry fixed effects, city fixed-effects and year fixed-effects are controlled. The regression is also clustered at the city level. *F*-test (*p*-value) describes the significance of the observations, and *F* is the statistical value of the *f*-test.

FE, fixed effect; ROA, return on total assets; SOE, state-owned enterprises.

*, significance at the 1% level; **, significance at the 5% level; ***, significance at the 10% level.

innovation in the firm in the current period. The second is the regression result of the high-dimensional fixed effect of managerial myopia on the number of green inventions authorised with a one-period lag. Regressions (1)~(4) show that there is indeed a significant inhibitory effect of managerial short-termism on the overall level of green innovation in the firm, with results significant at least at the 5% level. Regressions (5) to (8) show that management myopia has a very strong negative effect on the number of green invention patents granted with a lag of one period, and the results are significant at the 1% level. These results indicate that management short-sightedness does have a strong inhibitory effect on corporate green innovation, so the previous hypothesis H1 can be confirmed to be true in the static panel.

High-dimensional fixed-effects benchmark results – Dynamic panel

To further verify the robustness of the findings, Box 3 further examines the impact of management myopia on green innovation from a dynamic panel perspective. List 1 shows the regression results of management myopia on the number of green invention patent applications and grants, while List 2 shows the effect of management myopia on the overall level of green inventions and the number of non-contemporaneous green invention grants. The results show that under the dynamic panel, management myopia has a significant inhibitory effect on the number of green invention patent applications, the overall green innovation level and the number of green inventions granted in the preceding period at both the industry level and the firm level, and the findings are mostly significant at the 1% level. This further validates hypothesis H1 that management myopia has a significant negative effect on corporate green innovation.

Instrumental variable regression robustness test and external factors

Considering that the above model may still have endogeneity problems, this article continues to test the green innovation damaging effect of management myopia under static panel (Online Appendix 1 Table A1) and dynamic panel

BOX 2: Robustness testing: Lagged effects and overall comparison.

List	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables			**Green_sum**				**Green_grant**	
Myopia	-0.72***	-0.28***	-0.60***	-0.19**	-	-	-	-
	(-5.56)	(-3.00)	(-4.52)	(-2.10)	-	-	-	-
L_myopia	-	-	-	-	-0.34***	-0.19***	-0.30***	-0.15***
	-	-	-	-	(-3.54)	(-3.32)	(-2.97)	(-2.88)
Control	NO	NO	YES	YES	NO	NO	YES	YES
Cluster City	YES	YES	YES	YES	YES	YES	YES	YES
Company FEs	NO	YES	NO	YES	NO	YES	NO	YES
Industry FEs	YES	NO	YES	NO	YES	NO	YES	NO
City FEs	YES	YES	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES	YES	YES
N	48102	47598	48026	47527	42606	42207	42557	42162
R^2	0.22	0.66	0.34	0.67	0.15	0.59	0.24	0.60

Note: Robust standard errors in parentheses.
FEs, fixed effect.
**, $p < 0.05$; ***, $p < 0.01$.

BOX 3: Baseline regression 2.

List 1	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables			**Green_apply**				**Green_grant**	
Myopia	-0.29***	-0.18***	-0.21***	-0.10*	-0.12***	-0.07	-0.09*	-0.06
	(-6.57)	(-2.89)	(-4.53)	(-1.73)	(-2.71)	(-1.60)	(-1.89)	(-1.36)
L_ apply	0.766***	0.470***	0.720***	0.456***	-	-	-	-
	(42.01)	(27.35)	(46.89)	(27.10)	-	-	-	-
L_ grant	-	-	-	-	0.77***	0.48***	0.74***	0.47***
	-	-	-	-	(29.42)	(20.39)	(31.38)	(20.55)
N	42606	42207	42557	42162	42606	42207	42557	42162
R^2	0.65	0.71	0.66	0.72	0.60	0.68	0.61	0.68
List 2			**Green_sum**				**Green_grant**	
myopia	-0.29***	-0.19***	-0.22***	-0.12*	-	-	-	-
	(-6.03)	(-3.03)	(-4.08)	(-1.94)	-	-	-	-
L_green_sum	0.81***	0.51***	0.76***	0.49***	-	-	-	-
	(52.62)	(31.50)	(57.24)	(31.27)	-	-	-	-
L_myopia	-	-	-	-	-0.14***	-0.14***	-0.12***	-0.12***
	-	-	-	-	(-4.09)	(-3.74)	(-3.22)	(-3.08)
L_green_grant	-	-	-	-	0.77***	0.48***	0.74***	0.47***
	-	-	-	-	(29.40)	(20.39)	(31.34)	(20.54)
N	42606	42207	42557	42162	42606	42207	42557	42162
R^2	0.70	0.76	0.71	0.76	0.60	0.68	0.61	0.68
Control	NO	NO	YES	YES	NO	NO	YES	YES
Cluster City	YES	YES	YES	YES	YES	YES	YES	YES
Company FEs	NO	YES	NO	YES	NO	YES	NO	YES
Industry FEs	YES	NO	YES	NO	YES	NO	YES	NO
City FEs	YES	YES	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES	YES	YES

Note: This Box 3 reports the use of the same multidimensional fixed-effects and cluster order as the previous Box 2. Robust standard errors in parentheses.
FEs, fixed effect.
*, $p < 0.1$; ***, $p < 0.01$.

(Box 4), respectively, using the 2SLS method. This article regresses management myopia with a first-order lag as an instrumental variable and still uses high-dimensional fixed-effects. The results in Table 1-A1 and Box 4 indicate that management myopia is indeed pervasive and significant at both industry and firm levels for green innovation.

Furthermore, considering the influence of China's green policies, particularly the 'Air Pollution Prevention Action Plan' of 2013, which has had the most significant impact in the green domain, we divided the sample into two periods: 2001–2012 and 2013–2021, corresponding to before and after the policy implementation. The results (Online Appendix 1 Table A2) remain significant at the industry level across both periods, demonstrating that managerial short-termism consistently inhibits green innovation regardless of the policy shift. This robustness suggests that the negative effect of managerial myopia on green innovation is not solely driven by the policy shock but is a persistent phenomenon influenced by firm-level strategic decisions. The consistency across these periods further reinforces the reliability of our findings, highlighting the enduring challenge of short-termism in the context of evolving environmental regulations.

BOX 4: Heterogeneity test.

Standard 1: Firm nature								
List 1a	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Myopia		**Green_grant**				**Green_apply**		
SOEs	-0.15***	-0.01	-0.10**	-0.01	-0.25***	-0.04	-0.16**	-0.03
	(-3.47)	(-0.21)	(-2.17)	(-0.19)	(-3.21)	(-0.56)	(-2.21)	(-0.43)
Non-SOEs	-0.12	-0.15**	-0.05	-0.17*	-0.38***	-0.33***	-0.21**	-0.14*
	(-1.60)	(-2.360)	(-0.69)	(-1.90)	(-4.35)	(-3.39)	(-2.55)	(-1.65)
List 1b	**(9)**	**(10)**	**(11)**	**(12)**	**(13)**	**(14)**	**(15)**	**(16)**
Myopia		**Green_sum**				**F_green_grant**		
SOEs	-0.26***	-0.06	-0.18**	-0.05	-0.48***	-0.09	-0.34***	-0.09
	(-3.31)	(-0.76)	(-2.35)	(-0.62)	(-5.22)	(-1.15)	(-3.31)	(-1.07)
Non-SOEs	-0.37***	-0.35***	-0.20**	-0.17*	-0.33*	-0.30***	-0.18	-0.23***
	(-4.01)	(-3.47)	(-2.25)	(-1.82)	(-1.78)	(-3.50)	(-1.05)	(-2.96)
Standard 2: Heavy and non-heavy pollution enterprise								
List 2a	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Myopia		**Green_grant**				**Green_apply**		
Heavy pollution	-0.06	0.01	-0.06	-0.00	-0.23**	-0.10	-0.19**	-0.08
	(-0.80)	(0.10)	(-0.84)	(-0.00)	(-2.40)	(-0.10)	(-2.16)	(-0.82)
Non-heavy pollution ~	-0.15***	-0.11**	-0.11*	-0.08	-0.32***	-0.21***	-0.22***	-0.10
	(-2.70)	(-2.03)	(-1.81)	(-1.42)	(-5.82)	(-2.64)	(-3.74)	(-1.34)
List 2b	**(9)**	**(10)**	**(11)**	**(12)**	**(13)**	**(14)**	**(15)**	**(16)**
Myopia		**Green_sum**				**F_green_grant**		
Heavy pollution	-0.19*	-0.05	-0.16*	-0.04	-0.15**	-0.14*	-0.15**	-0.14*
	(-1.96)	(-0.50)	(-1.70)	(-0.35)	(-2.00)	(-1.83)	(-2.25)	(-1.87)
Non-heavy pollution ~	-0.34***	-0.25***	-0.25***	-0.14*	-0.14***	-0.14**	-0.11*	-0.10*
	(-6.24)	(-3.20)	(-4.00)	(-1.94)	(-2.71)	(-2.56)	(-1.79)	(-1.67)
Standard 3: CEO duality								
List 3a	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Myopia		**Green_grant**				**Green_apply**		
CEO Duality=1	-0.06	-0.00	-0.00	0.03	-0.27	-0.20	-0.14	-0.05
	(-0.38)	(-0.04)	(-0.02)	(0.28)	(-1.63)	(-1.49)	(-0.89)	(-0.38)
CEO Duality=0	-0.13***	-0.06	-0.10***	-0.06	-0.30***	-0.13*	-0.21***	-0.09
	(-3.98)	(-1.20)	(-2.67)	(-1.12)	(-5.55)	(-1.94)	(-3.85)	(-1.33)
List 3b	**(9)**	**(10)**	**(11)**	**(12)**	**(13)**	**(14)**	**(15)**	**(16)**
Myopia		**Green_sum**				**F_green_grant**		
CEO Duality=1	-0.28	-0.19	-0.15	-0.03	-0.19	-0.15*	-0.13	-0.11
	(-1.53)	(-1.40)	(-0.83)	(-0.25)	(-1.31)	(-1.77)	(-0.93)	(-1.30)
CEO Duality=0	-0.31***	-0.15**	-0.22***	-0.10	-0.14***	-0.11**	-0.11***	-0.09**
	(-5.71)	(-2.13)	(-3.94)	(-1.55)	(-3.79)	(-2.46)	(-2.76)	(-1.99)
Lag DV	YES	YES	YES	YES	YES	YES	YES	YES
Control	NO	NO	YES	YES	NO	NO	YES	YES
FEs and Cluster	YES	YES	YES	YES	YES	YES	YES	YES

Note: This Box 4 reports use of the same multidimensional fixed-effectsand cluster order as the previous Box 3. In each regression, the dependent variable (DV) is lagged one period and put into the regression term. Robust standard errors in parentheses.

SOE, state-owned enterprises; DV, dependent variable; FE, fixed effect; CEO, chief executive officer.

*, $p < 0.1$; **, $p < 0.05$; ***, $p < 0.01$.

Heterogeneity test

This article further analyses the heterogeneity of this influence path based on the former content to verify that management short-termism inhibits green innovation. This article discusses heterogeneity from the three dimensions of enterprise nature (whether it is a state-owned enterprise), pollution level (whether it is a heavily polluting enterprise) and the CEO duality (whether the chairman and CEO are the same person). The heterogeneous conclusions under these three dimensions are illustrated in Box 4. These results are derived from sample-split regressions, where separate analyses were conducted for each subsample based on the respective dimensions to ensure a rigorous and precise assessment of the heterogeneity.

Standard 1 results reveal that managerial myopia in non-state-owned enterprises (non-SOEs) exerts a stronger and more significant inhibitory effect on green innovation than in SOEs, with impacts evident at both firm and industry levels. This broader influence in non-SOEs aligns with Chinese policies mandating SOEs to prioritise green innovation, suggesting non-SOE managers must exercise greater caution against short-termism's detrimental effects on sustainability strategies. Standard 2 findings indicate that managerial short-termism more significantly hampers green innovation in non-heavily polluting firms, implying these firms' managers should be particularly vigilant, especially as ESG ratings increasingly tie green innovation to secondary market performance and market capitalisation. Standard 3 results

show that managerial myopia more severely undermines green innovation in non-dual-role firms. This may reflect dual-role firms' structural resistance to short-sighted policies or the enhanced ability of non-dual-role CEOs and chairs to enact short-termism, necessitating a more balanced approach in non-dual-role firms to safeguard green innovation's market competitiveness.

Mediating effect

Based on the H2 hypothesis, this section analyses the mediating effects of R&D investment (rdi), digital transformation level (digi) and ESG rating (esg), respectively. Box 5 reports the effect of managerial short-termism on the three mediating variables under different high-dimensional fixed-effects and the possible mediating effect on green innovation. List 1 controls for the industry fixed-effects, while List 2 fixes the firm level. The results show that managerial myopia has a direct negative effect on firms' R&D investment and digital transformation at both the firm and industry levels, and the results are significant at the 1% level.

Among the mediating effects at the industry level, the indirect effect of management myopia on green innovation through firms' R&D investment and digital transformation is highly significant, and the results are also significant at the 1% level. This indicates that management myopia at the industry level indirectly inhibits firms' green innovation through R&D investment and digital transformation, while the mediating effect at the firm level is not significant. The effect of management myopia on ESG ratings is not significant in the current period, but the effect of both on green innovation at the industry level is significant. Combined with the fact that the ESG coefficient is significantly positive at the 1% level, this suggests that ESG ratings also have a positive effect on green innovation. Given that there is a certain time lag phenomenon in ESG evaluation, it is difficult for management myopia to directly influence ESG rating in the present. Therefore, this article conducts a high-dimensional fixed-effects regression on management short-sightedness by front-loading the ESG rating to one period.

Further results show that managerial myopia has a significant inhibitory effect on ESG in the next period, whether at the industry level or at the firm level. Therefore, it can be shown that ESG ratings have a mediating effect on the green innovation-inhibiting effect of management myopia. In this section, the number of green patent applications is compared with the overall level of green innovation, and the difference in the results is not significant. The article also compares the results of industry fixed-effects and firm fixed-effects results separately and presents more objective regression results by using a more stringent model setting. All these initiatives increase the robustness of the conclusions in this article.

Conclusion

This study provides robust causal evidence on the detrimental impact of managerial myopia on green innovation, leveraging high-dimensional fixed-effects models and textual analysis of Management Discussion and Analysis (MD&A) from Chinese listed companies (2001–2021). The findings reveal that managerial myopia significantly suppresses green innovation at both firm and industry levels, with pronounced effects in non-state-owned enterprises (non-SOEs), non-heavy polluting firms and non-CEO-duality firms. Mediation analysis identifies reduced R&D investment, limited digital transformation and weaker ESG management as critical mechanisms through which short-termism undermines sustainable innovation. These results contribute to the literature by integrating MD&A-based measures of myopia with patent-based green innovation metrics, offering a novel perspective on corporate governance and environmental sustainability in an emerging market context.

The scientific contribution of this research lies in its methodological innovation and its advancement of corporate governance and sustainability scholarship. By combining textual analysis of MD&A with quantitative patent data, the study captures both managerial intent and innovation outcomes, providing a nuanced understanding of how short-termism disrupts long-term environmental goals. Unlike prior studies that primarily focus on financial performance, this research elucidates the mediating roles of R&D investment, digital transformation and ESG management in the relationship between managerial myopia and green innovation. Additionally, by examining heterogeneity across firm types, it offers contextual insights into the factors shaping myopia's effects, enriching the theoretical framework for studying corporate sustainability in emerging markets.

The practical implications of this study extend beyond policy recommendations, offering actionable strategies for corporate leaders to counter managerial myopia and foster green innovation. Firms can integrate ESG metrics into strategic decision-making to align managerial incentives with sustainability goals, particularly in non-SOEs where myopia's effects are more pronounced. Adopting long-term equity incentives can shift managerial focus from short-term profits to sustained innovation, while proactive digital transformation can streamline processes and support data-driven sustainability strategies. Furthermore, strengthening ESG management practices, such as incorporating green innovation into performance appraisals, can enhance firm reputation and investor confidence, as ESG ratings increasingly influence market valuations. These actions not only mitigate the adverse effects of short-termism but also position firms as leaders in sustainable business practices, creating competitive advantages in markets prioritising environmental responsibility.

To address managerial myopia and promote green innovation, this study proposes several policy recommendations. Firstly, firms should enhance managers' awareness of ESG values to foster a sustainability-oriented mindset. Secondly, optimising incentive structures with long-term equity incentives can align

BOX 5: Mediation model.

List 1	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
Variables	**rdi**	**apply**	**sum**	**digi**	**Apply**	**sum**	**esg**	**apply**	**sum**
myopia	-0.65***	-0.32***	-0.34***	-0.53***	-0.17***	-0.17***	-0.01	-0.25***	-0.25***
	(-4.66)	(-3.18)	(-2.96)	(-7.00)	(-3.63)	(-3.22)	(-0.70)	(-3.91)	(-3.53)
rdi	-	0.075***	0.075***	-	-	-	-	-	-
	-	(20.06)	(21.67)	-	-	-	-	-	-
digi	-	-	-	-	0.03***	0.03***	-	-	-
	-	-	-	-	(8.82)	(9.06)	-	-	-
esg	-	-	-	-	-	-	-	0.13***	0.11***
	-	-	-	-	-	-	-	(5.64)	(4.81)
L_rdi	0.81***	-	-	-	-	-	-	-	-
	(101.86)	-	-	-	-	-	-	-	-
L_digi	-	-	-	0.75***	-	-	-	-	-
	-	-	-	(110.41)	-	-	-	-	-
L_esg	-	-	-	-	-	-	0.70***	-	-
	-	-	-	-	-	-	(70.52)	-	-
L_g_apply	-	0.66***	-	-	0.72***	-	-	0.71***	-
	-	(47.09)	-	-	(46.17)	-	-	(45.52)	-
L_g_sum	-	-	0.71***	-	-	0.76***	-	-	0.75***
	-	-	(56.31)	-	-	(56.45)	-	-	(56.50)
_cons	-0.03	-0.80***	-0.73***	0.52***	-0.45***	-0.41***	0.26***	-0.63***	-0.56***
	(-0.21)	(-6.45)	(-5.50)	(5.59)	(-6.48)	(-5.58)	(8.81)	(-7.52)	(-6.15)
Control	YES	YES	YES	YES	YES	YES	YES	YES	YES
City and Year Fes	YES	YES	YES	YES	YES	YES	YES	YES	YES
Industry FEs	YES	YES	YES	YES	YES	YES	YES	YES	YES
N	24860	26786	26786	42557	42557	42557	31764	33680	33680
R^2	0.89	0.65	0.70	0.82	0.66	0.71	0.58	0.65	0.70
List 2	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
Variables	**rdi**	**apply**	**sum**	**digi**	**Apply**	**sum**	**esg**	**apply**	**Sum**
myopia	-0.53***	-0.01	-0.06	-0.42***	-0.07	-0.09	-0.02	0.04	0.03
	(-2.718)	(-0.08)	(-0.42)	(-4.44)	(-1.29)	(-1.51)	(-1.17)	(0.44)	(0.31)
Rdi	-	0.09***	0.10***	-	-	-	-	-	-
	-	(11.02)	(12.52)	-	-	-	-	-	-
digi	-	-	-	-	0.04***	0.04***	-	-	-
	-	-	-	-	(10.23)	(10.27)	-	-	-
Esg	-	-	-	-	-	-	-	0.21***	0.19***
	-	-	-	-	-	-	-	(8.16)	(7.38)
L_rdi	0.51***	-	-	-	-	-	-	-	-
	(29.86)	-	-	-	-	-	-	-	-
L_digi	-	-	-	0.52***	-	-	-	-	-
	-	-	-	(62.92)	-	-	-	-	-
L_esg	-	-	-	-	-	-	0.44***	-	-
	-	-	-	-	-	-	(38.09)	-	-
L_green_apply	-	0.28***	-	-	0.45***	-	-	0.33***	-
	-	(21.81)	-	-	(27.28)	-	-	(24.33)	-
L_green_sum	-	-	0.29***	-	-	0.49***	-	-	0.36***
	-	-	(24.19)	-	-	(31.48)	-	-	(28.27)
_cons	2.43***	-1.13***	-0.96***	1.06***	-0.47***	-0.38***	0.91***	-0.89***	-0.72***
	(8.17)	(-5.06)	(-3.97)	(6.33)	(-4.88)	(-3.63)	(24.14)	(-6.12)	(-4.64)
Control	YES	YES	YES	YES	YES	YES	YES	YES	YES
City and Year FEs	YES	YES	YES	YES	YES	YES	YES	YES	YES
Company FEs	YES	YES	YES	YES	YES	YES	YES	YES	YES
N	24418	26355	26355	42162	42162	42162	31384	33286	33286
R^2	0.90	0.74	0.78	0.85	0.72	0.76	0.67	0.74	0.78

Note: Robust standard errors in parentheses.
RDI, research & development investment; digi, digital transformation level; ESG, environmental, social and governance; FE, fixed effect.
***, $p < 0.01$.

managerial decisions with green innovation goals. Thirdly, companies should incorporate green metrics into performance appraisal systems to reward sustainable practices. Fourthly, strengthening green information disclosure by mandating comprehensive reporting of green innovation and ESG performance can enhance transparency and stakeholder trust. Fifthly, firms should proactively promote digital transformation to support long-term innovation strategies, ensuring managers prioritise sustainable technologies and opportunities over short-term gains.

This study is not without limitations, which warrant acknowledgement to contextualise its findings. The reliance on MD&A textual analysis may introduce managerial bias, as disclosures may not fully reflect true intentions. The focus on Chinese listed companies limits the generalisability of the findings to other markets with different regulatory and cultural contexts. In addition, the use of patent-based measures for green innovation may exclude non-patentable innovations, such as process improvements, potentially underestimating firms' sustainability efforts. Finally, the mediation analysis assumes linear relationships, which may oversimplify the complex interactions between managerial myopia, R&D investment, digital transformation and ESG management.

Future research can build on this study by addressing its limitations and exploring new directions. Incorporating alternative data sources, such as employee surveys or third-party ESG audits, could validate MD&A-based measures of managerial myopia. Extending the analysis to other emerging and developed markets would provide insights into cross-country variations in myopia's impact on green innovation. Developing broader metrics for green innovation, including non-patentable outcomes such as circular economy practices, could offer a more comprehensive view of sustainability efforts. Employing non-linear models or machine learning techniques could better capture the complex mediating mechanisms involved. Finally, investigating the role of external stakeholders, such as institutional investors or regulators, in mitigating managerial myopia's effects could further enrich the understanding of corporate sustainability.

In conclusion, this study underscores the critical need to address managerial myopia to foster green innovation, offering actionable insights for firms and policymakers. By integrating ESG values, optimising incentives and embracing digital transformation, companies can overcome short-termism to achieve sustainable development. The findings highlight the business value of green innovation, not only for environmental outcomes but also for long-term profitability and stakeholder trust. As sustainability becomes a global imperative, this research provides a roadmap for aligning corporate governance with environmental and economic goals, contributing to a more resilient and responsible business landscape.

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Competing interests

The authors declare that they have no financial or personal relationships that may have inappropriately influenced them in writing this article.

CRediT authorship contribution

Mingkun Zhou: Data curation, Formal analysis, Investigation, Methodology, Resources, Visualisation, Writing-review & editing. Simin Yang: Conceptualisation, Formal analysis, Software, Validation. Xuhua Qiu: Investigation, Project administration, Visualisation, Writing-original draft. Yun Zeng: Formal analysis, Investigation, Validation, Visualisation. Kunjie Zhu: Data curation, Formal analysis, Investigation, Methodology, Resources, Validation, Visualisation, Writing-original draft. All authors reviewed the article, contributed to the discussion of results, approved the final version for submission and publication, and take responsibility for the integrity of its findings.

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Data availability

The data that support the findings of this study are available from the corresponding author, Kunjie Zhu, upon reasonable request.

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